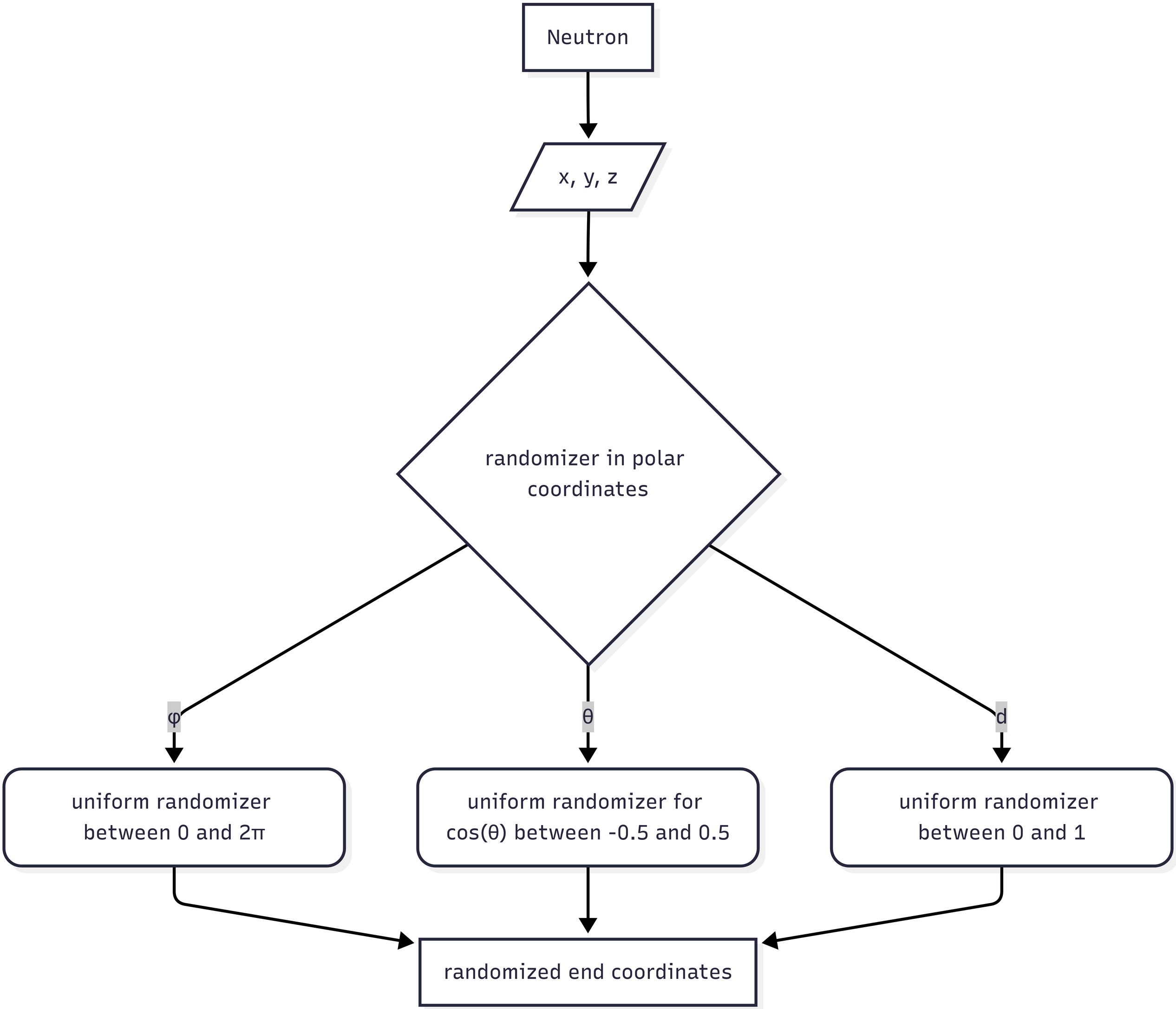

Neutron
x, y, z
randomizer in polar coordinates
φ
θ
d
uniform randomizer between 0 and 2π
uniform randomizer for cos(θ) between -0.5 and 0.5
uniform randomizer between 0 and 1
randomized end coordinates